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Intelligent Multimodal approach for Automatic Generation of High-Impact Short Formats (Shorts / Reels)

Prepared by

Ben Amor Hanine
Bhar Amira
Cheriaa Nermine

Defended on june 2026 in front of a jury composed of:

……	President of the jury
Mliki Hazar	Supervisor at INSAT

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Acknowledgments

Abstract

Write your abstract here.

Keywords: ..., ...

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State of the Art

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3 Fusion-Based Multimodal Systems

4 Joint Multimodal Representation Learning

Joint multimodal representation learning addresses a fundamental limitation shared by both unimodal and fusion-based approaches: the absence of a unified semantic space in which modalities such as vision, audio, and text can be collectively treated and compared. Rather than processing each signal independently and combining outputs,the joint approach trains a single model to map all modalities into a shared high-dimensional em-

bedding space, where geometric proximity reflects semantic similarity regardless of the originating modality.

Modern multimodal systems are inspired by the contrastive learning framework popularized by CLIP-like architectures, enabling cross-modal reasoning where heterogeneous modalities are treated as different expressions of the same underlying objective. Joint multimodal representation learning addresses...

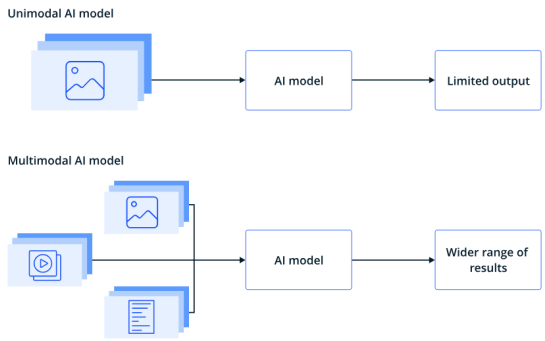

Figure 1.1: Comparison of unimodal and multimodal processing frameworks. Unimodal systems process a single data type, while multimodal systems integrate multiple modalities into a unified representation [AI Planet, 2024].

4.1 Architectural Overview

Although joint multimodal models vary in design, most share a common three-stage architecture: modality-specific encoders, a fusion mechanism, and optionally a decoder for generative tasks Sajid [2024].

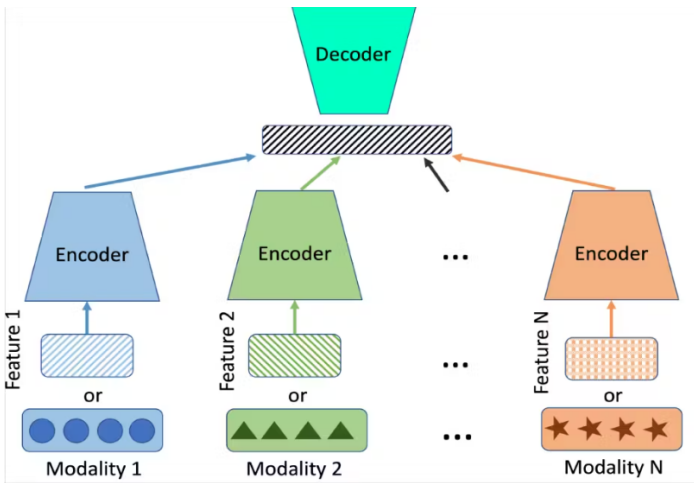

Figure 1.2: Joint multimodal model Architecture
Sajid [2024]

Modality-specific encoders transform raw inputs into dense feature vectors: image

encoders employ **CNNs** or **Vision Transformers (ViT)**, text encoders leverage **transformer architectures** to map token sequences into contextual embeddings, and audio encoders such as Wav2Vec2 [Baevski et al., 2020] convert raw waveforms into representations capturing rhythm, tone, and semantic content. Once encoded, a fusion mechanism integrates these embeddings into a unified representation. Four strategies are commonly used: *early fusion* combines raw features before processing; *intermediate fusion* merges latent representations; *late fusion* combines final modality outputs; and *hybrid fusion* applies elements of all three at different stages. These strategies are implemented through attention-based methods **?**, feature concatenation, or dot-product interactions between modality vectors.

4.2 Contrastive Learning Framework

Contrastive learning is the fundamental training strategy underlying modern multimodal models. Rather than predicting explicit class labels, the model learns by comparing pairs of inputs: a **positive pair** consists of semantically aligned samples from different modalities — an image and its corresponding caption, or a video frame and its paired audio segment — while a **negative pair** consists of mismatched samples. The model is trained to pull positive pairs closer together in the embedding space while pushing negative pairs apart, with the supervision signal arising entirely from cross-modal correspondence rather than manual annotation.

This is formalized through the InfoNCE objective [van den Oord et al., 2019]. Given a batch of N matched pairs $(x_i^{(A)}, x_i^{(B)})$ from two modalities A and B, each encoder maps its input to a **normalized** embedding such that the correct match is identifiable among all N candidates. The symmetric loss is:

$$\mathcal{L} = -\frac{1}{2N} \sum_{i=1}^{N} \left[\log \frac{e^{\mathrm{sim}(\mathbf{e}_i^A, \mathbf{e}_i^B)/\tau}}{\sum_j e^{\mathrm{sim}(\mathbf{e}_i^A, \mathbf{e}_j^B)/\tau}} + \log \frac{e^{\mathrm{sim}(\mathbf{e}_i^B, \mathbf{e}_i^A)/\tau}}{\sum_j e^{\mathrm{sim}(\mathbf{e}_i^B, \mathbf{e}_j^A)/\tau}} \right] \tag{1.1}$$

where $\mathrm{sim}(\cdot, \cdot)$ is cosine similarity and τ is a temperature hyperparameter controlling the sharpness of the similarity distribution. A lower τ sharpens the objective, producing a more discriminative but noise-sensitive embedding space; in practice, τ is either fixed — CLIP uses $\tau = 0.07$ [**?**] — or learned per modality, as in ImageBind [Girdhar et al., 2023]. This framework has proven highly scalable, enabling multimodal pretraining on large-scale web data without any manually labelled supervision.

4.3 Shared Embedding Spaces

The product of contrastive training is a shared embedding space, typically of dimension $d \in \{512, 768, 1024\}$, in which each modality is processed by a dedicated encoder — a transformer-based language encoder for text, a vision transformer for images, or a spectrogram-based encoder for audio [Sajid, 2024] — followed by a projection layer that maps the resulting features into the same latent dimensionality. All embeddings are constrained to lie on the unit hypersphere $\mathbb{S}^{d-1}$ via ℓ_2 **normalization**, making cosine similarity a universal proximity measure across modalities:

$$\text{sim}(\mathbf{e}^A, \mathbf{e}^B) = \mathbf{e}^A \cdot \mathbf{e}^B \qquad \text{since } \|\mathbf{e}\| = 1 \tag{1.2}$$

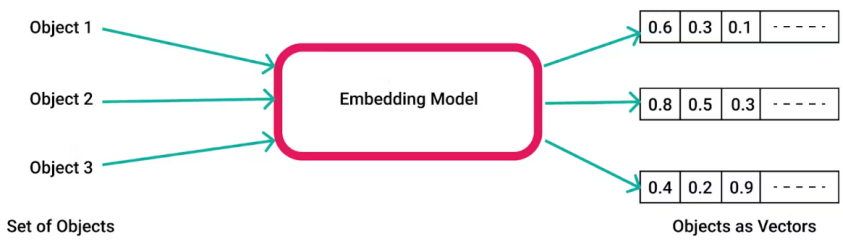

Figure 1.3: Multimodal embeddings: raw inputs from different modalitiesprojected into a common high-dimensional vector space Sajid [2024].

Within this shared space, semantically related inputs are positioned close to one another regardless of modality: an image of a cheering crowd, an audio clip of applause, and a textual description of excitement occupy nearby regions of the embedding manifold. This unified representation facilitates cross-modal similarity computation and retrieval without requiring explicit modality-specific fusion mechanisms.

A further property is *emergent transitivity*: a model trained on image-audio and image-text pairs separately achieves implicit audio-text alignment through the shared visual anchor, without the two modalities ever being trained together directly.
ImageBind [Girdhar et al., 2023] exploits this property to bind six modalities — image, text, audio, depth, thermal, and IMU — using only image-paired data for each, achieving full cross-modal alignment across all modality combinations at inference time.

4.4 Zero-Shot Transfer and Cross-Modal Retrieval

The shared embedding space directly enables zero-shot transfer: the ability to perform recognition or scoring on novel concepts expressed as natural language, without any task-specific training. Because the model learns semantic alignment rather than task-specific boundaries, it can transfer to unseen categories without fine-tuning. A concept such as "crowd applause" is encoded by the text encoder into a semantic anchor embedding; candidate segments are then scored by their cosine similarity to this anchor, regardless of modality:

$$\text{score}(x) = \text{sim}(f_{\text{text}}(q),\ f_{\text{target}}(x)) \tag{1.3}$$

The shared space enables cross-modal retrieval: a natural language query can retrieve the most semantically relevant video segment purely on embedding proximity, while an audio clip of crowd cheering can retrieve the visual moment of a goal.
This "search by anything" capability supports text-to-image, image-to-text, audio-to-text, and audio-to-video retrieval within a single framework.

4.5 Advantages and Limitations of Joint Multimodal Learning

Joint multimodal learning paradigm offers four key advantages over alternative approaches.It provides **native cross-modal alignment**, to encode semantic correspondences directly in the geometry of the embedding space rather than through engineered fusion rules. It enables **zero-shot generalization**, allowing the system to handle new content categories at inference time through natural language, without depending on annotated datasets. It supports **modality-agnostic scoring**, where a single similarity function operates uniformly across all available modalities.

Finally, models (such as ImageBind) benefit from **large-scale pretraining** on internet-scale paired data, encoding world knowledge that transfers effectively to downstream tasks without fine-tuning.

Despite these strengths, the paradigm presents notable limitations. The most significant is the **absence of temporal modeling**: joint multimodal models produce a single embedding per fixed-length segment and cannot model sequential dynamics across segments. Narrative structures, such as the comedic buildup across several exchanges before a punchline, must be handled by external temporal analysis. A second limita-

tion is **embedding norm uniformity**: because all embeddings are ℓ_2-normalized, their magnitude carries no information, making raw energy-based scoring inapplicable without additional statistical processing. Finally, **modality imbalance** in pretraining data: image-text pairs, being far more abundant than audio-centric combinations, can reduce the discriminative power of audio embeddings in content where acoustic cues are the primary signal.

Addressing these challenges remains an active area of research, particularly for video-centric applications where temporal coherence and multimodal interaction play a central role.

5 Representative Multimodal Models

5.1 CLIP

5.2 CLAP

Contrastive Language-Audio Pretraining (CLAP), as described in [Elizalde et al., 2022], is an extension of the contrastive learning paradigm of CLIP to the audio domain, with the aim of learning a joint representation space between audio and text to facilitate zero-shot audio classification and retrieval tasks. The architecture employs two encoders to process the audio and text modalities in the form of mel spectrogram and text, respectively, and is trained using symmetric contrastive loss to maximize the similarity between matching and minimize the similarity between non-matching audio and text pairs.

Despite its success in aligning the audio and text modalities, the limitations of the CLAP approach for the task of highlight detection in videos are as follows: the audio is processed in isolation without access to the visual information, thus making it impossible to leverage the audio-visual modalities. In addition, the encoders are optimized for short videos, thus limiting the modeling of long-range dependencies, which is critical in the case of videos. The use of architectures such as ImageBind, which combines the three modalities in a single representation space, can be more effective for multimodal video analysis.

5.3 Flamingo

The Flamingo model developed by DeepMind is a powerful family of Visual Language Models designed to process sequences of images or videos interleaved with text and generate free-form text as output.
Flamingo represents a notable progression in vision-language modeling by enabling a single model to perform multiple multimodal tasks with minimal task-specific fine-tuning.

A core concept of its architecture is the fusion of large, pre-trained, and frozen models specifically a **vision encoder** and a **language model**to leverage their existing knowledge without the need for intensive retraining .
To efficiently bridge these two modalities, Flamingo introduces **the Perceiver Resampler**, which compresses a huge number of visual features into a fixed-size collection of visual tokens , these tokens then condition the frozen language model through **Gated Cross-Attention layers**, which are interleaved between the original language model layers , these gated layers are essential for preserving the stability of the model during training and ensuring the language model's original performance remains intact at initialization.

A key contribution of Flamingo is its **few-shot learning** capability The model can adapt to various tasks with only a few input-output examples . Experimental evaluations showed strong performance across several visionlanguage benchmarks, such as multimodal reasoning tasks, visual question answering and image captioning .

5.4 LLAVA

LLaVA is a multimodal model designed to enhance general-purpose visual and language understanding through the use of visual instruction tuning

The architecture of LLaVA consists of a **pretrained CLIP vision encoder** and a **Vicuna large language model**, which are connected by a two-layer **Multi-Layer Perceptron cross-modal projector** .
This MLP projector converts visual encoder tokens into post-embedding layer LLM tokens, replacing the simpler linear projection used in earlier versions to improve multimodal capabilities .
The training process follows a **two-stage pipeline:** an initial stage focused on modality

alignment using roughly 600,000 image-text pairs, followed by a second stage of visual instruction fine-tuning where the model is trained on diverse conversation datasets .
This approach enables LLaVA to perform tasks such as image description, visual question answering, and multimodal dialogue generation. Compared to more complex architectures, LLaVA is considered more accessible and computationally efficient while maintaining competitive performance.

5.5 Video-LLaMA

Video-LLaMA [Zhang et al., 2023] is a video understanding model that uses the generative method of incorporating video content into large language models. Unlike embedding-based video retrieval models that treat text, vision, and audio as equivalent modalities in an embedding space, Video-LLaMA places the language model at the core of the model and treats vision and audio as conditionals for text generation. The model is instruction-tuned to process video content with synchronized audio for natural language reasoning on dynamic video content.

The model employs two branches: the Vision-Language branch that aggregates video features using Video Q-Former to extract temporal dependencies across frames using cross-frame attention mechanisms, and the Audio-Language branch that processes audio content using ImageBind [Girdhar et al., 2023] to extract features that can be used in conjunction with the language model.
Although Video-LLaMA is effective in processing temporal video content for scene understanding and question answering tasks, its autoregressive generation method is not appropriate for embedding-based video retrieval tasks.

However, the training of video language models like Video-LLaMA is computationally expensive and demands large-scale video-text paired datasets, which limits its adoption by many research groups.

VideoLLaMA 2 [Cheng et al., 2024] is an improved version of VideoLLaMA that uses a spatial-temporal convolution connector instead of the dual branches of the original model to link the vision and audio models to the language model and improves audio content processing using BEATs [Chen et al., 2022].

5.6 ImageBind

6 Comparative Analysis and Discussion

7 Positioning of the Proposed Approach

Conclusion

Conclusion and perspectives

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Appendix 1

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References

Bibliography

AI Planet. Transfer learning and multimodal models. AI Planet LLM Bootcamp, Module 9, 2024. Image credit: LeewayHertz. Available at: https://aiplanet.com/learn/llm-bootcamp/module-9/2366/transfer-learning-and-multimodal-models.

Alexei Baevski, Henry Zhou, Abdelrahman Mohamed, and Michael Auli. wav2vec 2.0: A framework for self-supervised learning of speech representations, 2020. URL https://arxiv.org/abs/2006.11477.

Sanyuan Chen, Yu Wu, Chengyi Wang, Shujie Liu, Daniel Tompkins, Zhuo Chen, and Furu Wei. Beats: Audio pre-training with acoustic tokenizers, 2022. URL https://arxiv.org/abs/2212.09058.

Zesen Cheng, Sicong Leng, Hang Zhang, Yifei Xin, Xin Li, Guanzheng Chen, Yongxin Zhu, Wenqi Zhang, Ziyang Luo, Deli Zhao, and Lidong Bing. Videollama 2: Advancing spatial-temporal modeling and audio understanding in video-llms. *arXiv preprint arXiv:2406.07476*, 2024.

Benjamin Elizalde, Soham Deshmukh, Mahmoud Al Ismail, and Huaming Wang. Clap: Learning audio concepts from natural language supervision, 2022. URL https://arxiv.org/abs/2206.04769.

Rohit Girdhar, Alaaeldin El-Nouby, Zhuang Liu, Mannat Singh, Kalyan Vasudev Alwala, Armand Joulin, and Ishan Misra. Imagebind: One embedding space to bind them all, 2023. URL https://arxiv.org/abs/2305.05665.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Clip learning transferable visual models from natural language supervision. 2021. URL https://arxiv.org/abs/2103.00020.

Haziqa Sajid. Top 10 multimodal models. Encord Blog, 2024. Available at: https://encord.com/blog/top-multimodal-models/.

Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding, 2019. URL https://arxiv.org/abs/1807.03748.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, pages 5998–6008, 2017.

Hang Zhang, Xin Li, and Lidong Bing. Video-llama: An instruction-tuned audio-visual language model for video understanding. *arXiv preprint arXiv:2306.02858*, 2023.